

Module: **Large Language Models**

Travel planning agent

1. Problem and Scenario

Planning a trip in real life can be a time-consuming and inconvenient process because travelers often need to gather information from several different sources. For example, a user may need to search for suitable transportation, compare prices based on their budget and number of passengers, check weather conditions, find suitable hotels, and identify places to visit at the destination. These activities are usually performed separately using different websites, applications, and services, requiring the user to manually collect and compare the information.

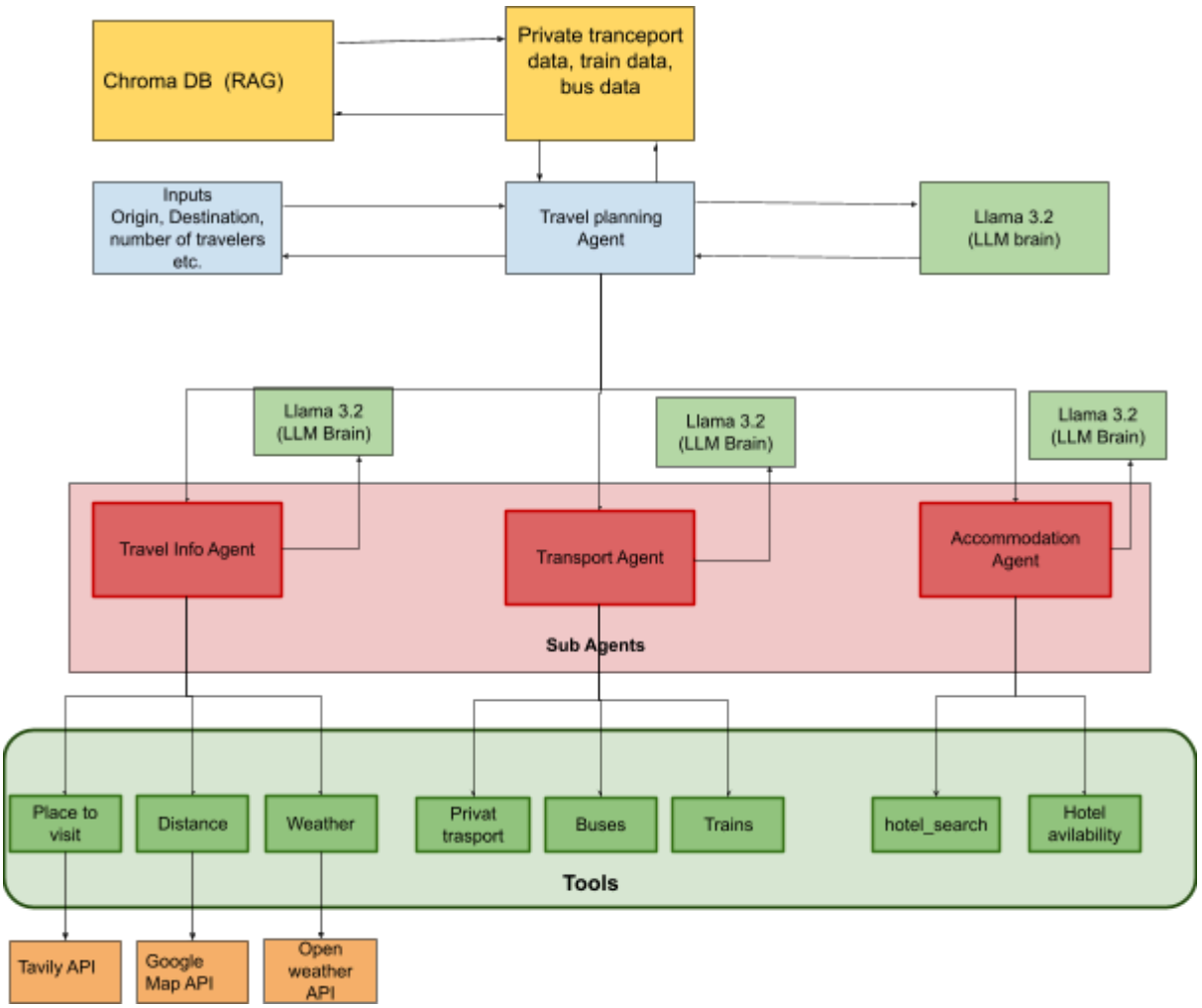
Another challenge is that travel decisions are often dependent on each other. For example, the choice of transportation may depend on the number of passengers, available budget, travel distance, and weather conditions. Similarly, the number of days in a trip can influence the required accommodation and the number of places that can be visited. Managing these requirements manually can therefore make the planning process more complicated and time-consuming.

To address these challenges, the Travel Planning Agent was developed as an intelligent system that assists users in creating structured travel plans according to their requirements. Instead of requiring users to search for each piece of information separately, the agent can coordinate multiple tools and information sources to support the planning process.

The user can provide requirements such as the origin, destination, travel date, number of passengers, number of days, travel type, and transportation budget. Based on these requirements, the system can provide suitable transportation information, including private transportation and public transportation options such as buses and trains. The system can also retrieve weather information for the destination and identify places to visit. Hotel and accommodation information can be incorporated to support multi-day trips.

The main objective of the Travel Planning Agent is therefore to reduce the manual effort required for trip planning by combining different travel-related services into a single intelligent system. By using an agent-based architecture, the system can determine which tools are required for a particular request and combine the retrieved information to produce a structured and user-oriented travel plan.

2. System architecture



3. Agents

The Travel Planning Agent uses a multi-agent architecture in which different travel-related responsibilities are assigned to specialized sub-agents. This approach separates the functionality into smaller components, allowing each sub-agent to focus on a specific area of the travel-planning process. The system consists of three main sub-agents: the **Travel Information Agent**, **Transport Agent**, and **Accommodation Agent**.

3.1 Travel Information Agent

The Travel Information Agent is responsible for providing general information about the user's destination. It can determine the distance between the origin and destination, retrieve weather information for a specified location and date, and identify places that users can visit.

The agent has access to specialized tools such as the distance tool, weather tool, and places-to-visit tool. For example, when a user requests information about travelling from Colombo to Kandy, the Travel Information Agent can obtain the distance between the two

locations and retrieve relevant weather information for the planned travel date. It can also provide recommendations for attractions and places of interest in Kandy.

This agent therefore acts as a specialized information provider for the main Travel Planning Agent rather than directly generating the complete travel itinerary.

3.2 Transport Agent

The Transport Agent is responsible for identifying suitable transportation options according to the user's travel requirements. It considers factors such as the origin, destination, number of passengers, travel budget, and required travel information.

The system supports different transportation categories, including private transportation, public buses, and trains. Private transportation information is retrieved from a dedicated transportation knowledge base using a Retrieval-Augmented Generation (RAG) system. Similarly, public bus and train information can be maintained in separate knowledge bases and retrieved through their respective RAG systems.

The Transport Agent can therefore compare available transportation information and provide options that are appropriate for the user's requirements. For example, if a user is travelling from Colombo to Kandy with a limited budget, the agent can retrieve relevant bus or other transportation information and provide suitable options.

3.3 Accommodation Agent

The Accommodation Agent is responsible for finding suitable accommodation based on the user's travel requirements. It considers factors such as the destination, number of passengers, number of days, accommodation preferences, and budget.

The agent can search for hotels or other suitable accommodation options and provide relevant information to the main Travel Planning Agent. Where booking functionality is supported, the Accommodation Agent can also assist with the hotel-booking process according to the user's selected requirements.

By separating accommodation-related tasks from transportation and general travel information, the system allows the Accommodation Agent to focus specifically on finding appropriate places to stay.

3.4 Interaction Between the Agents

The three specialized agents work together through the main Travel Planning Agent. The main agent acts as the coordinator, interpreting the user's request and determining which specialized agent or tool is required.

For example, for a multi-day trip from Colombo to Kandy, the main agent may obtain distance and weather information from the Travel Information Agent, transportation recommendations from the Transport Agent, and accommodation options from the Accommodation Agent. The information returned by these components can then be combined to produce a structured travel plan that matches the user's requirements.

This multi-agent design improves modularity because each sub-agent has a clearly defined responsibility. It also makes the system easier to maintain and extend, since additional travel-related capabilities can be added without significantly changing the responsibilities of the existing agents.

4. Tools

Tools are an important component of the Travel Planning Agent because they allow the agents to access external services and project-specific knowledge that cannot be reliably provided by the language model alone. Each tool is designed to perform a specific task and return the required information to the agent. The main tools in the system include distance calculation, weather information, places-to-visit search, private transportation retrieval, public bus retrieval, train information retrieval, and accommodation-related services.

4.1 Distance Tool

The Distance Tool is responsible for obtaining the distance between the user's origin and destination. It receives two parameters: the origin and destination. The tool uses a mapping service to retrieve the route distance and related travel information.

For example, when the user requests a journey from Colombo to Kandy, the tool can be called with Colombo as the origin and Kandy as the destination. The returned distance is then provided to the relevant agent and can be used when evaluating transportation options.

4.2 Weather Tool

The Weather Tool retrieves weather information for a specified location and date. It uses the OpenWeather API to obtain forecast information and returns details such as temperature, weather conditions, and probability of rain.

The tool receives the location and date as input. The date is converted into the required `YYYY-MM-DD` format before requesting the forecast. This information can be useful when selecting transportation or planning outdoor activities. For example, if heavy rain is expected on the planned travel date, the agent can consider this information when providing recommendations.

4.3 Places-to-Visit Tool

The Places-to-Visit Tool is responsible for identifying attractions and points of interest at the user's destination. It allows the Travel Information Agent to retrieve

destination-specific information instead of relying only on the language model's existing knowledge.

The tool can be used when the user wants recommendations for places to visit. The retrieved information can then be incorporated into the final travel plan based on the user's destination and number of travel days.

4.4 Private Transport Tool

The Private Transport Tool provides private transportation recommendations based on the user's requirements. It receives information such as the origin, destination, number of passengers, and transportation budget.

The tool uses a Retrieval-Augmented Generation (RAG) system to retrieve relevant information from the project's private transportation knowledge base. The knowledge base is stored in `transport_data.txt`, while the `transport_rag.py` module handles document loading, text splitting, embedding generation, and retrieval using Chroma.

The retrieved transportation information is combined with the route distance obtained through the Distance Tool. This allows the main agent to provide transportation options that are relevant to the user's route, group size, and budget.

4.5 Bus Tool

The Public Bus Tool provides information about public bus transportation. It follows a similar RAG architecture to the Private Transport Tool but uses a separate knowledge base specifically for bus information.

The bus information is stored in `bus_data.txt` and processed through `bus_rag.py`. The RAG system uses Ollama embeddings and Chroma to retrieve relevant bus information based on the user's requested route and requirements.

For example, for a Colombo-to-Kandy journey, the tool can retrieve relevant route numbers, operators, bus types, approximate travel times, frequency, and other information stored in the bus knowledge base.

4.6 Train Tool

The Train Tool is responsible for retrieving information about railway transportation. Similar to the Public Bus Tool, it can use a dedicated RAG knowledge base containing train-related information.

The tool can retrieve relevant information such as train routes, available services, travel times, classes, and other information contained in the train knowledge base.

Separating train information into its own knowledge base allows the system to maintain and update railway information independently from bus and private transportation information.

4.7 Hotel Search Tool

The Hotel Search Tool is responsible for finding hotels near the user's requested destination. It receives the destination as an input and uses the Wavely API to retrieve relevant hotel information in that location.

For example, when the user requests a trip to Kandy, the Accommodation Agent can provide the destination to the Hotel Search Tool. The tool then uses the Wavely API to search for nearby hotels and return

5. Agent Loop

The loop begins when the user provides a travel request. The Main Travel Planning Agent analyzes the request and identifies the information required to satisfy the user's needs. Based on the request, it determines which specialized agent or tool should be used.

For example, if the user requests a trip from Colombo to Kandy, the Main Agent may require information about the route distance, weather conditions, transportation, places to visit, and accommodation. The Main Agent can therefore delegate these tasks to the appropriate components.

The Travel Information Agent handles general destination-related information. It can use the Distance Tool to determine the distance between the origin and destination, the Weather Tool to retrieve weather conditions for the travel date, and the Places-to-Visit Tool to identify attractions at the destination.

The Transport Agent handles transportation-related requirements. Depending on the user's requirements, it can retrieve private transportation information from the private transport RAG system or retrieve public bus and train information from their respective knowledge bases. The agent considers requirements such as the number of passengers, destination, route, and transportation budget when providing suitable options.

The Accommodation Agent handles accommodation-related tasks. It can use the Hotel Search Tool to retrieve hotel information near the destination through the Wavely API. The returned information can then be evaluated according to requirements such as the number of passengers, duration of the trip, destination, and accommodation budget.

After a tool or specialized agent is executed, the result is returned to the Main Travel Planning Agent. The Main Agent evaluates the returned information and determines whether additional information or another tool is required. If more information is required, the agent continues the loop by selecting the next appropriate tool or specialized agent. If all required information has been collected, the Main Agent combines the retrieved results and generates the final structured travel plan.

The overall process can therefore be represented as:

User Request → Analyze → Select Agent/Tool → Execute → Receive Result → Evaluate → Select Next Agent/Tool or Generate Final Plan.

6. Prompt Design

Prompt design is used to guide the behavior of the Travel Planning Agent and its specialized agents. Each agent is given a system prompt that defines its role, responsibilities, available tools, and limitations.

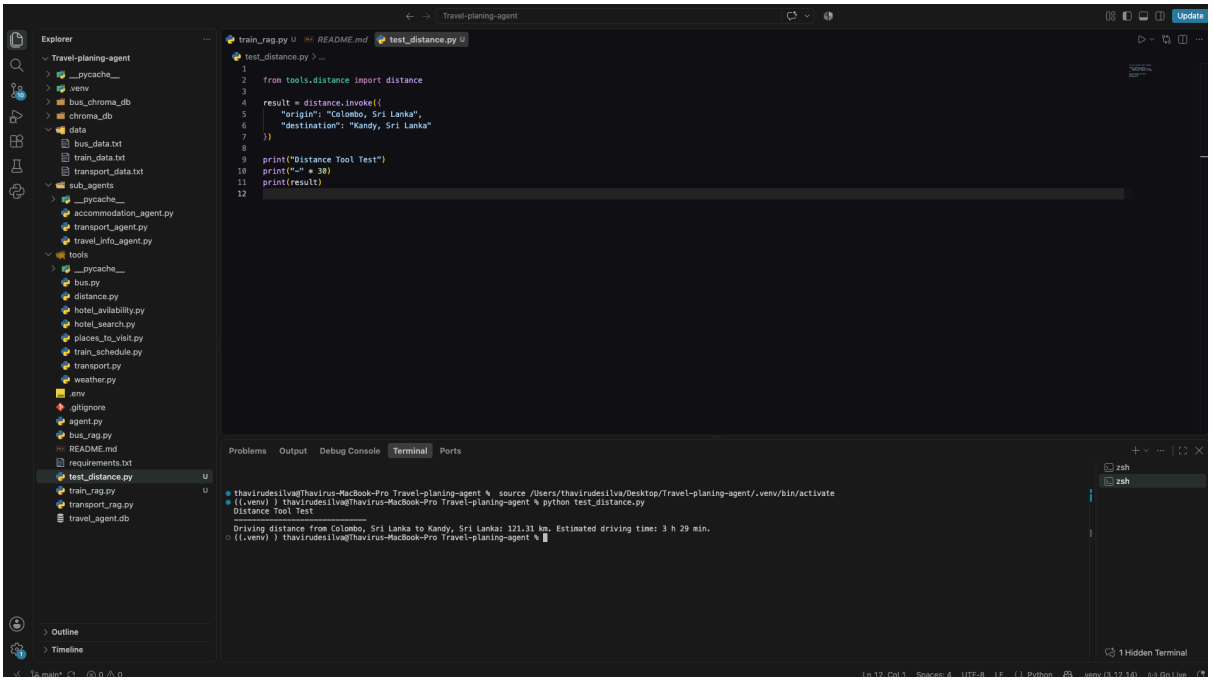
The Main Travel Planning Agent prompt instructs the agent to understand the user's travel requirements, coordinate the specialized agents, select the appropriate tools, and combine the retrieved information into a structured travel plan.

The Travel Information Agent prompt defines its responsibility for retrieving distance, weather, and places-to-visit information. It is also instructed to use the Weather Tool only when a valid location and date are provided.

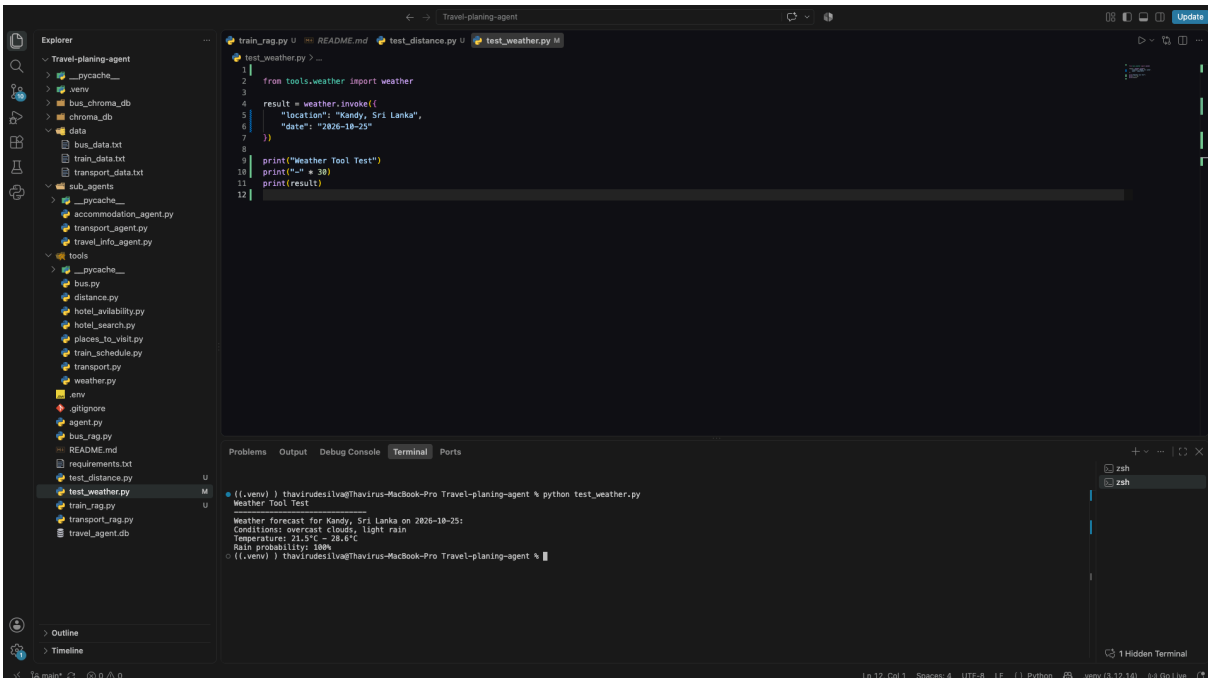
The Transport Agent prompt focuses on finding suitable transportation based on factors such as origin, destination, passengers, and budget. The Accommodation Agent focuses on finding suitable hotels based on the user's requirements.

7. Testing

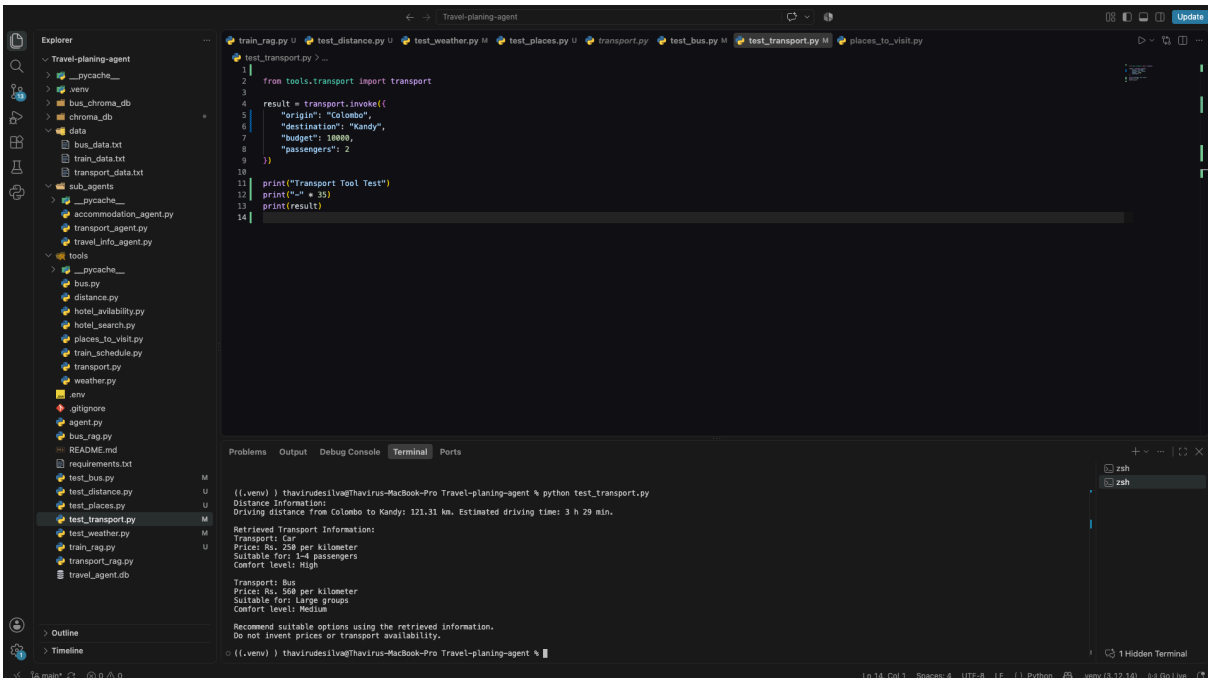
7.1. Distance tool testing



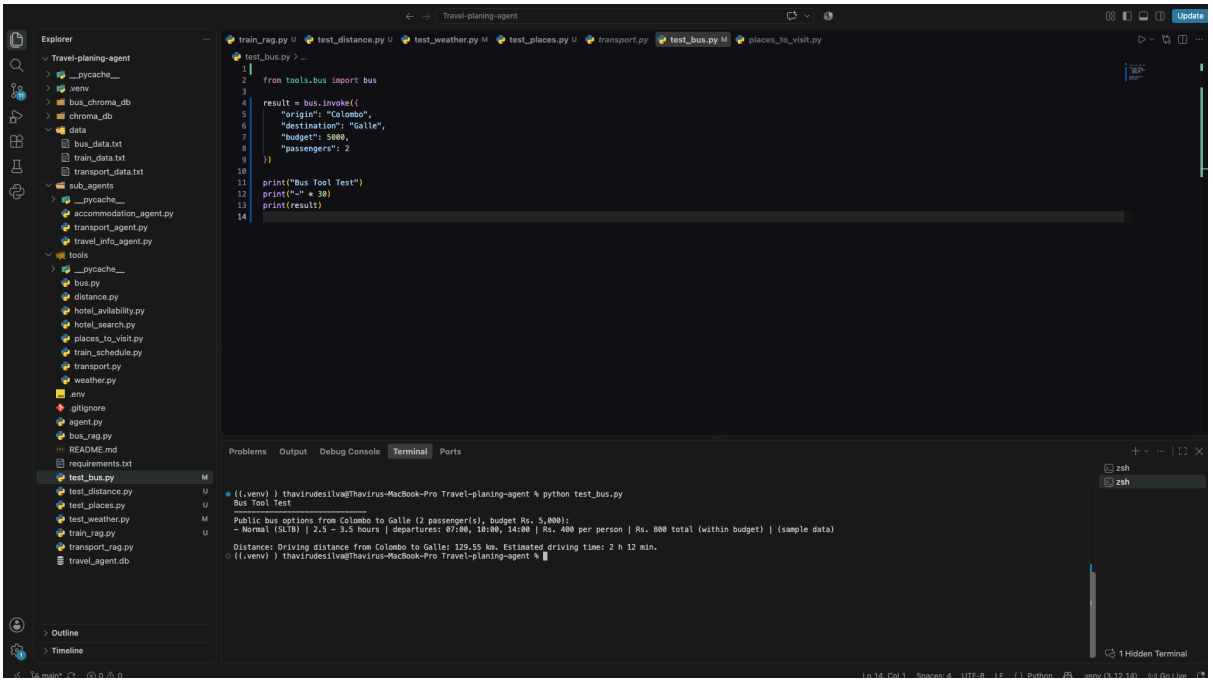
7.2. Weather tool testing



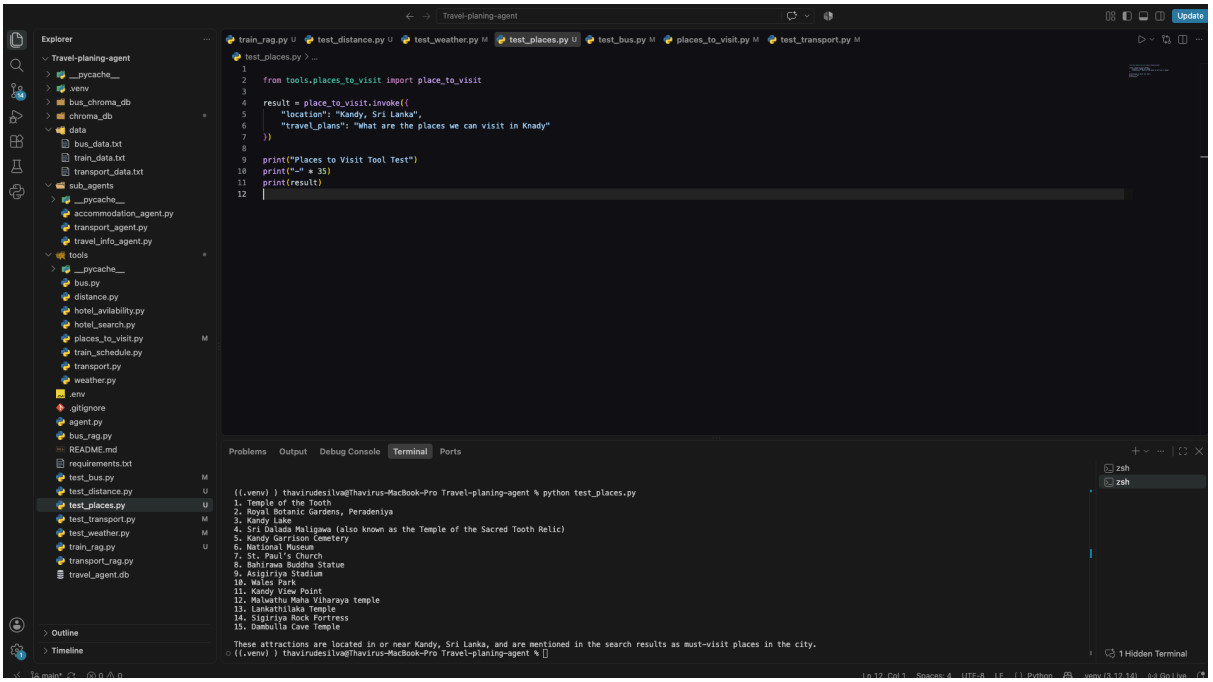
7.3. Privet transport tool testing



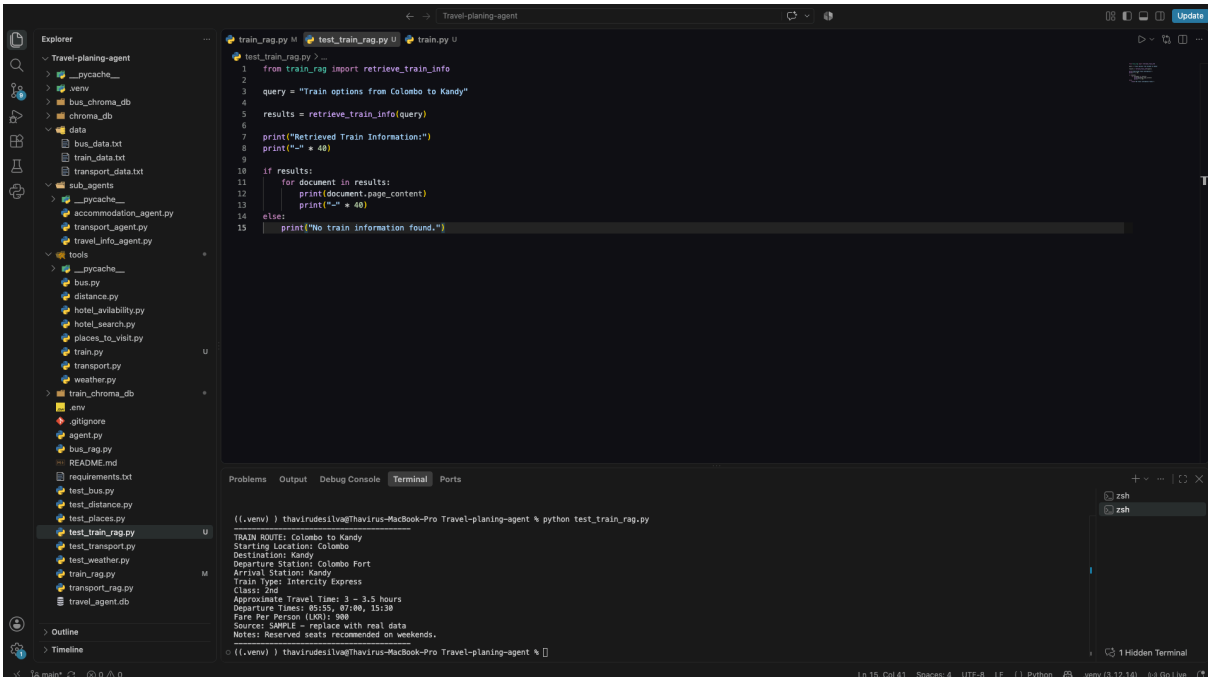
7.4. Public bus tool testing



7.5. Places tool testing



7.6. Train tool testing



8. Limitations

The Travel Planning Agent can automate several parts of the travel-planning process, the current system has some limitations.

- **Data dependency:** The accuracy of recommendations depends on the quality and freshness of the RAG knowledge bases and external APIs.
- **Limited real-time information:** Some transportation and accommodation information may not always represent real-time availability, prices, or schedules.
- **API dependency:** Weather, distance, and hotel-related features depend on external APIs. API failures, invalid API keys, or service interruptions can affect the system.
- **Limited booking functionality:** The current system mainly provides travel and accommodation information. Full real-time booking and payment functionality may not be available.
- **LLM limitations:** The language model may occasionally misunderstand user requirements or select an inappropriate tool, despite the instructions provided in the system prompts.
- **Knowledge-base coverage:** The RAG system can only provide reliable information for routes, transportation types, and other information that has been included in its knowledge bases.
- **Scalability:** As more transportation routes, hotels, and travel information are added, maintaining and updating the knowledge bases becomes more complex.

9. Future Improvements

Several improvements can be introduced to enhance the functionality of the Travel Planning Agent in the future.

- **Hotel Booking and Meal Planning:** Implement hotel booking functionality by considering the number of families or travelers, calculating the required number of rooms, and collecting preferences such as budget and hotel type. The system could also ask users about breakfast, lunch, and dinner requirements to provide more personalized accommodation and meal plans.
- **International Travel Support:** Expand the system beyond Sri Lanka to support international travel. This would include searching for flight schedules and prices, assisting with flight bookings, finding hotels at international destinations, and recommending attractions in different countries.
- **Real-Time Information:** Integrate more reliable real-time data sources to provide updated transportation schedules, fares, flight prices, hotel availability, and weather forecasts.

- **Personalized Itineraries:** Improve the system to generate detailed day-by-day itineraries based on the user's budget, travel duration, interests, group size, and preferred activities.

These improvements would help transform the current travel-planning assistant into a more comprehensive platform capable of supporting personalized domestic and international travel.

10. Individual contributions

Thaviru De Silva	Transport RAG, Train RAG, main Agent, distance tool, weather tool, bus tool, train tool, public transport tool, place_to_visit tool and Repot
Dulaj	Hotel search tool, accomodation_agent, Bus RAG and Demo video